

Non-star maximizers for additive abelian Cayley completion

Xiaojian Zeng

PBC School of Finance, Tsinghua University, 100190

xiaojianzeng@foxmail.com

Abstract

The additive abelian Cayley completion distance measures the number of edges needed to embed a graph as a spanning subgraph of an abelian Cayley graph of the same order, normalized by its original edge count. We disprove the conjecture that stars uniquely attain the connected-graph upper bound. For every odd $n \geq 5$, subdividing one edge of $K_{1,n-2}$ gives a non-star tree B_n with $\gamma^+(B_n) = (n-2)/2$. Every admissible host is K_n : its degree is at least $n-2$ and must be even. We also prove that five is the minimum counterexample order in the conjecture's range $n \geq 4$.

1 The invariant and the uniqueness question

All graphs are finite, simple and undirected. Let G be connected, with $n = |V(G)| \geq 4$ and $m = |E(G)|$. For an abelian group Γ of order n and a generating set $S = -S \subseteq \Gamma \setminus \{0\}$, the graph $\text{Cay}(\Gamma, S)$ has vertex set Γ , with x adjacent to $x + s$ for $s \in S$. Following Fokam Souop and Bitjoka [1, Definition 1], define

$$\gamma^+(G) = \frac{1}{m} \min_{\Gamma, S, \pi} (|E(\text{Cay}(\Gamma, S))| - m), \quad (1)$$

where $\pi : V(G) \rightarrow \Gamma$ is a bijection and $\pi(G)$ must be a spanning subgraph of $\text{Cay}(\Gamma, S)$. The minimization includes all such groups, generating sets and labelings. It is nonempty because $K_n = \text{Cay}(\mathbb{Z}/n\mathbb{Z}, (\mathbb{Z}/n\mathbb{Z}) \setminus \{0\})$ is an admissible host.

Completion to K_n and the inequality $m \geq n-1$ give

$$\gamma^+(G) \leq \frac{\binom{n}{2} - m}{m} = \frac{n(n-1)}{2m} - 1 \leq \frac{n-2}{2}. \quad (2)$$

For $K_{1,n-1}$, every regular spanning host has degree $n-1$, so the last bound is attained. This is the extremal bound of [1, Theorem 12]. The additive part of [1, Conjecture 13] asserts the implication

$$\gamma^+(G) = \frac{n-2}{2} \implies G \simeq K_{1,n-1}. \quad (3)$$

We refute (3) with an explicit family. The argument specializes the regularity and parity bound already present in [1, Theorem 5]; its contribution is the counterexample and its minimum order. For adjacent context on trees with bijective labels in finite abelian groups, see Zeng's rainbow-tree constructions [2]. That reference concerns a different labeling condition and is not used in the proof below.

2 The counterexamples and their minimum order

For $n \geq 4$, let B_n have vertex set $\{0, 1, \dots, n-1\}$ and edge set

$$E(B_n) = \{\{0, i\} : 1 \leq i \leq n-2\} \cup \{\{1, n-1\}\}. \quad (4)$$

Equivalently, B_n is obtained by subdividing one edge of $K_{1, n-2}$.

Theorem 1. *For every odd integer $n \geq 5$, B_n is a connected non-star tree, every admissible host in (1) is K_n , and*

$$\gamma^+(B_n) = \frac{n-2}{2}.$$

In particular, the uniqueness implication (3) is false.

Proof. The vertices $0, \dots, n-2$ form a star, and the remaining vertex is attached to vertex 1. Thus B_n is a tree with $n-1$ edges. Its degrees are $n-2$ at vertex 0, 2 at vertex 1, and 1 at each of the other $n-2$ vertices. In particular, it is not $K_{1, n-1}$.

Fix any admissible triple (Γ, S, π) , and write $H = \text{Cay}(\Gamma, S)$. The neighbors of each $x \in \Gamma$ are exactly the distinct vertices $x + s$, $s \in S$, so H is regular of degree $d = |S|$. Spanning containment and simplicity imply

$$n-2 \leq d \leq n-1.$$

The handshaking identity $nd = 2|E(H)|$ makes d even because n is odd. Since $n-2$ is odd, necessarily $d = n-1$. Hence $H = K_n$. This reasoning holds for every group and every labeling in the minimization.

Conversely, the cyclic complete host given after (1) contains B_n , for example under the identity labeling. The minimum is therefore attained, and

$$\gamma^+(B_n) = \frac{\binom{n}{2} - (n-1)}{n-1} = \frac{n-2}{2}.$$

□

At order five, the witness has edges 01, 02, 03, 14, where ij denotes $\{i, j\}$. Its degree sequence is $(3, 2, 1, 1, 1)$. Its only admissible host is K_5 , and the six edges to add are

$$04, \quad 12, \quad 13, \quad 23, \quad 24, \quad 34.$$

Consequently $\gamma^+(B_5) = 6/4 = 3/2$.

Proposition 2. *The minimum order of a counterexample to (3) in the range $n \geq 4$ is five.*

Proof. Let G be connected on four vertices. If $m \geq 4$, completion to K_4 gives $\gamma^+(G) \leq 6/m - 1 \leq 1/2 < 1$, whereas the target value is 1. If $m = 3$, then G is a tree. A vertex of degree three makes it the star; otherwise its maximum degree is at most two. A connected acyclic graph of maximum degree at most two is a path, so the only non-star possibility is P_4 . Adding the edge between its endpoints gives $C_4 = \text{Cay}(\mathbb{Z}/4\mathbb{Z}, \{1, 3\})$, whence $\gamma^+(P_4) \leq 1/3 < 1$. There is no counterexample at order four, and B_5 supplies one at order five. □

3 Exact checks and scope

The proofs above require no computation. As a finite consistency check, an exact enumeration for B_5 tests all symmetric generating sets in $\mathbb{Z}/5\mathbb{Z}$ and all 120 bijections. These are all group types, since a group of prime order five is cyclic. There are three generating sets, $\{1, 4\}$, $\{2, 3\}$ and $\{1, 2, 3, 4\}$, hence 360 set-labeling pairs. Only the complete host is feasible, for all 120 labelings, and each requires six additions.

A separate exact check enumerates every spanning supergraph of the four specified graphs B_4, B_5, B_6, B_7 , tests regularity by vertex degrees, and verifies explicit cyclic hosts attaining the resulting lower bounds. The respective enumeration sizes are 8, 64, 1024, 32768, totaling 33864. The corresponding values of γ^+ are $1/3, 3/2, 7/5, 5/2$. These computations use integer arithmetic and exact rational fractions. They concern four specified graphs, rather than a census of all graphs; the infinite odd-order result follows from Theorem 1. Scripts and exact outputs accompany the source package, with replay instructions in its README.

Section 7.1 of [1], version 1, reports confirmation of uniqueness through seven vertices. The witnesses B_5 and B_7 contradict that report under Definition 1 of the same version. We have not audited the source's census implementation or identified the cause of the discrepancy. The upper bound (2) remains valid. The present argument neither classifies all maximizers nor addresses the edit-distance part of the source conjecture. The literature comparison is restricted to the cited version and a bounded search on 7 September 2026; no claim of priority is made.

Computational and AI assistance. OpenAI Codex assisted with the counterexample argument, exact Python verification scripts, a separate-context mathematical review, and manuscript preparation. The review included a separate implementation and replays of the supplied checks. This assistance does not constitute external peer review. No proof assistant was used, and no computation replaces the proofs given here.

References

- [1] Rigobert Fokam Souop and Laurent Bitjoka. The Cayley completion of a graph. arXiv:2608.30894v1, 2026. Version 1, 31 August 2026.
- [2] Zijian Zeng. Coset and fiber constructions for rainbow trees in finite Abelian groups. SSRN preprint, 2026. doi:10.2139/ssrn.7387259.